Polarization of E M waves (light, in particular)

Plane electromagnetic wave – plane wave soln to Maxwell's wave eqn.

$$\vec{E} = \hat{x} E_{0x} \cos(\omega t - kz + \phi_0) \qquad \vec{B} = -\hat{y} \frac{E_{0x}}{c} \cos(\omega t - kz + \phi_0)$$

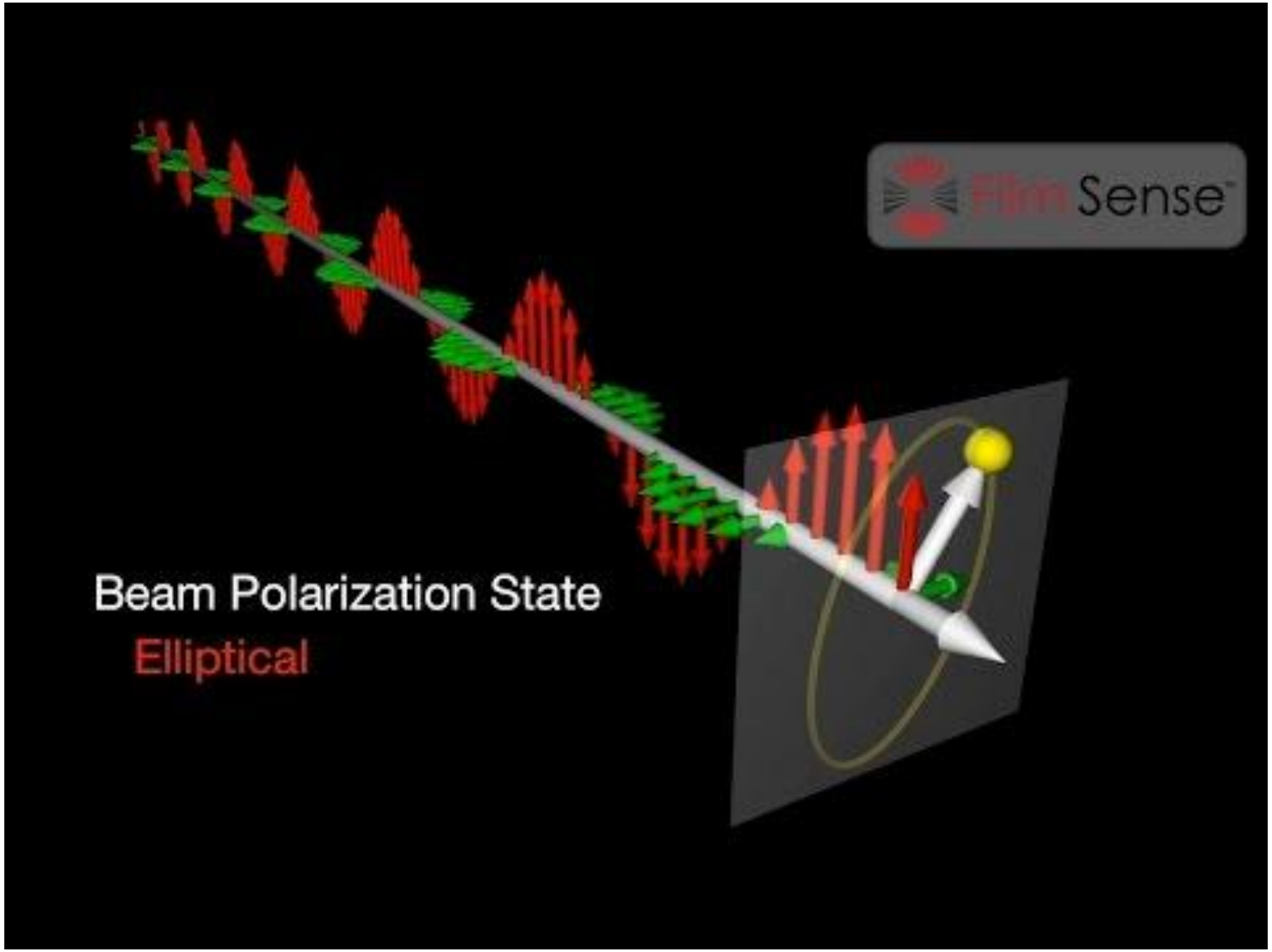
Nature of electric and magnetic fields of a plane E M wave

- Electric and magnetic fields are perpendicular to each other and both of them are perpendicular to the direction of propagation → E M waves are transverse in nature
- Electric and magnetic fields oscillate in phase
- Electric field strength is much stronger than the magnetic field strength,

$$\left|\vec{B}\right| = \frac{\left|\vec{E}\right|}{c}; \quad c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}; \quad \text{and} \quad \omega = kc$$

c is the speed of light in vacuum

Animated visuals of polarization of light



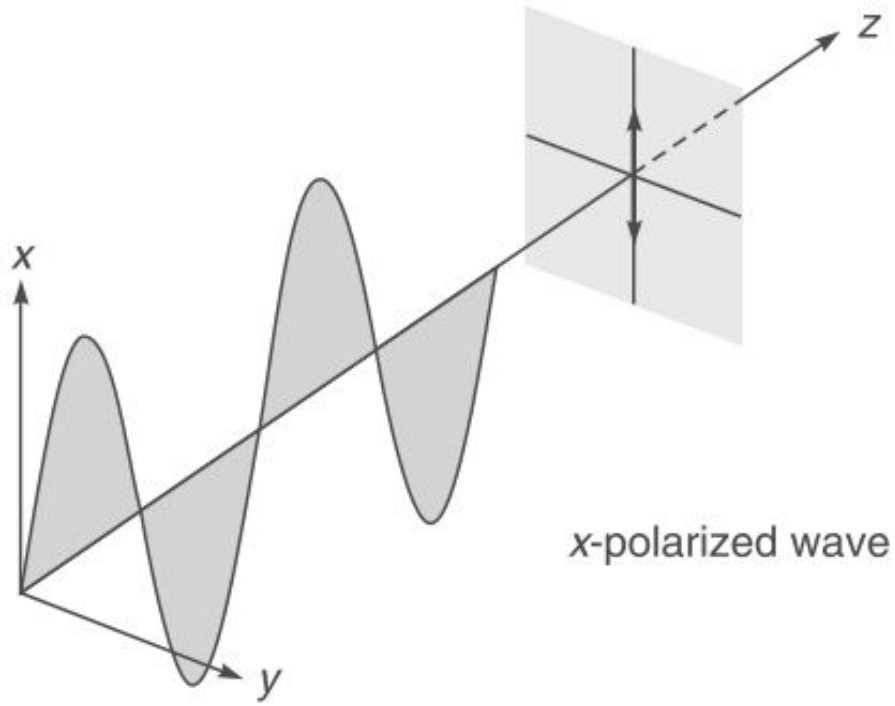
https://www.youtube.com/watch?v=BycPkRIutqg

Polarization of plane EM waves

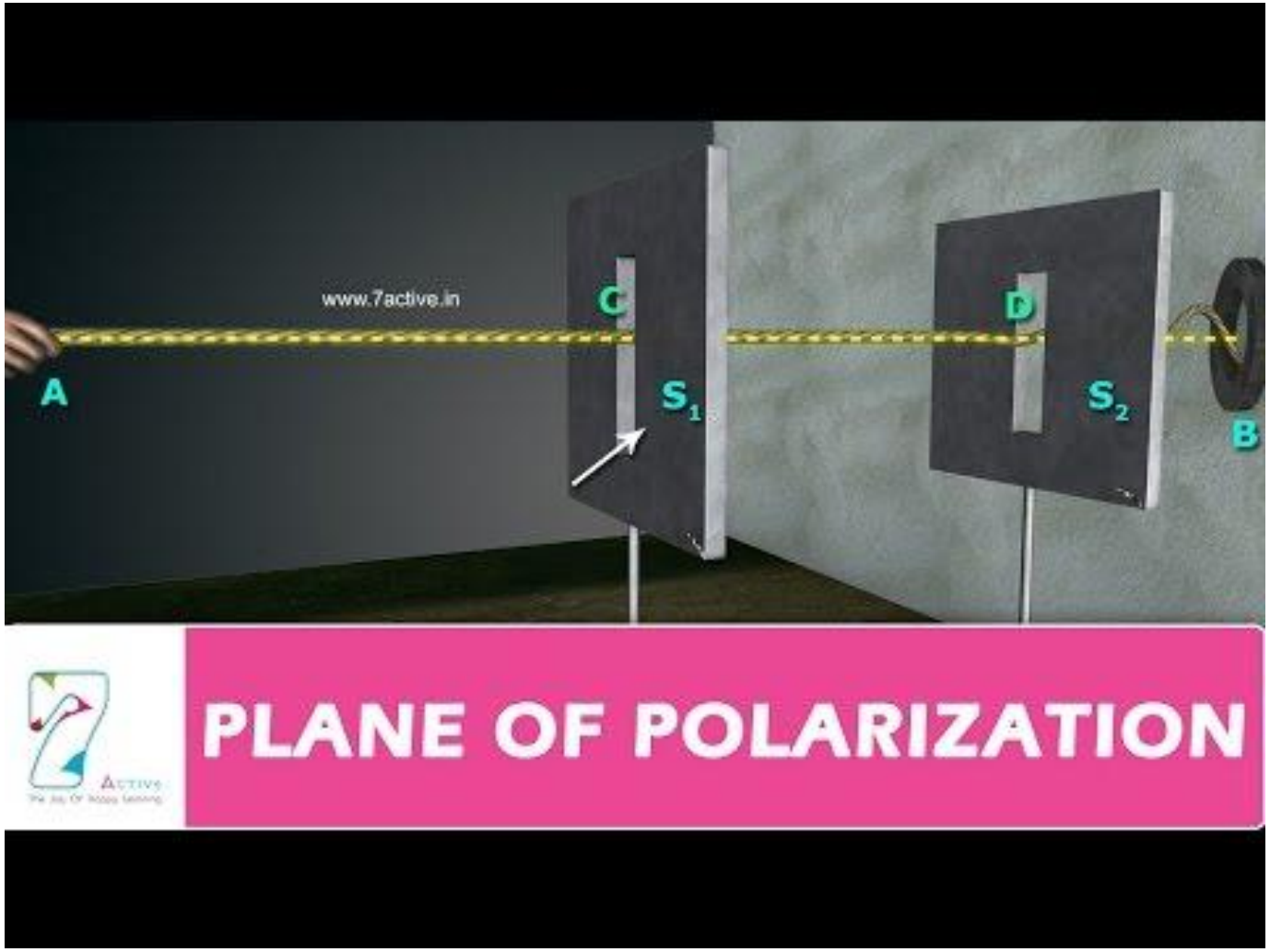
- Polarization of plane EM wave is decided by the electric field direction and its time evolution
- **For plane EM waves, polarization is defined by the projection of the trajectory (path traced in time) of the tip of the electric field vector in a plane perpendicular to the direction of propagation**

Polarization - analogy to transverse wave on a rope

- Imagine wave is generated on a rope by oscillating one end in a vertical plane
- Vibration exists only in one plane - in the vertical plane
- Tip of the amplitude (of vertical displacement) vector will trace a vertical line on a plane perpendicular to the direction of propagation
- This is an example of vertically linearly polarized wave



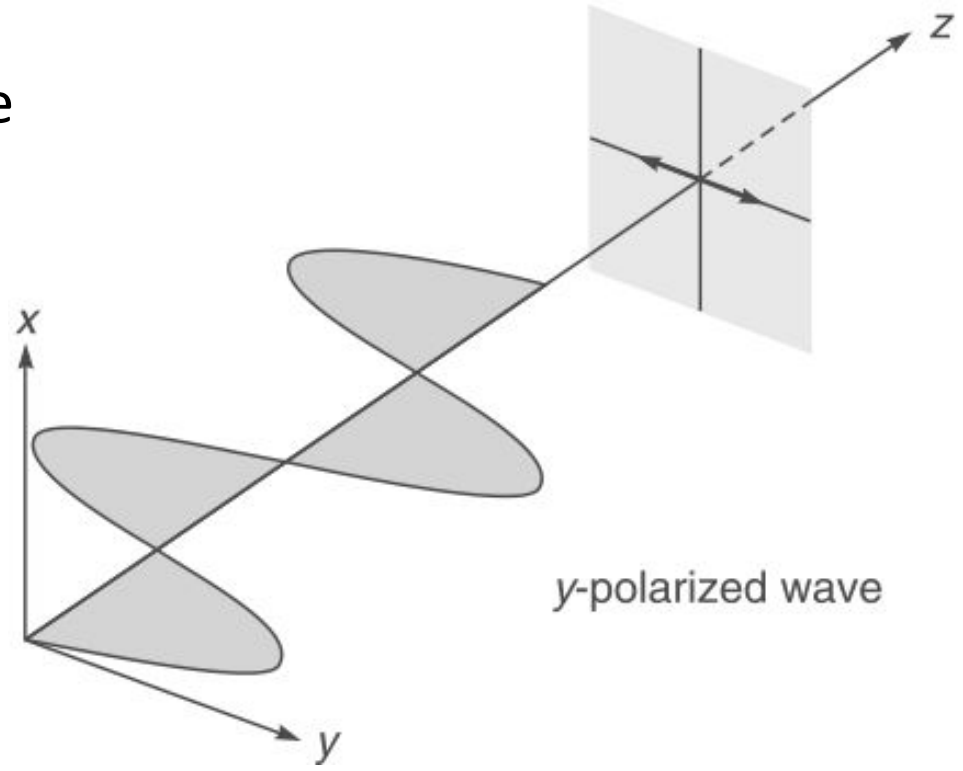
Vertically polarized transverse wave on a rope



https://www.youtube.com/watch?v=wP-HME8fpnI

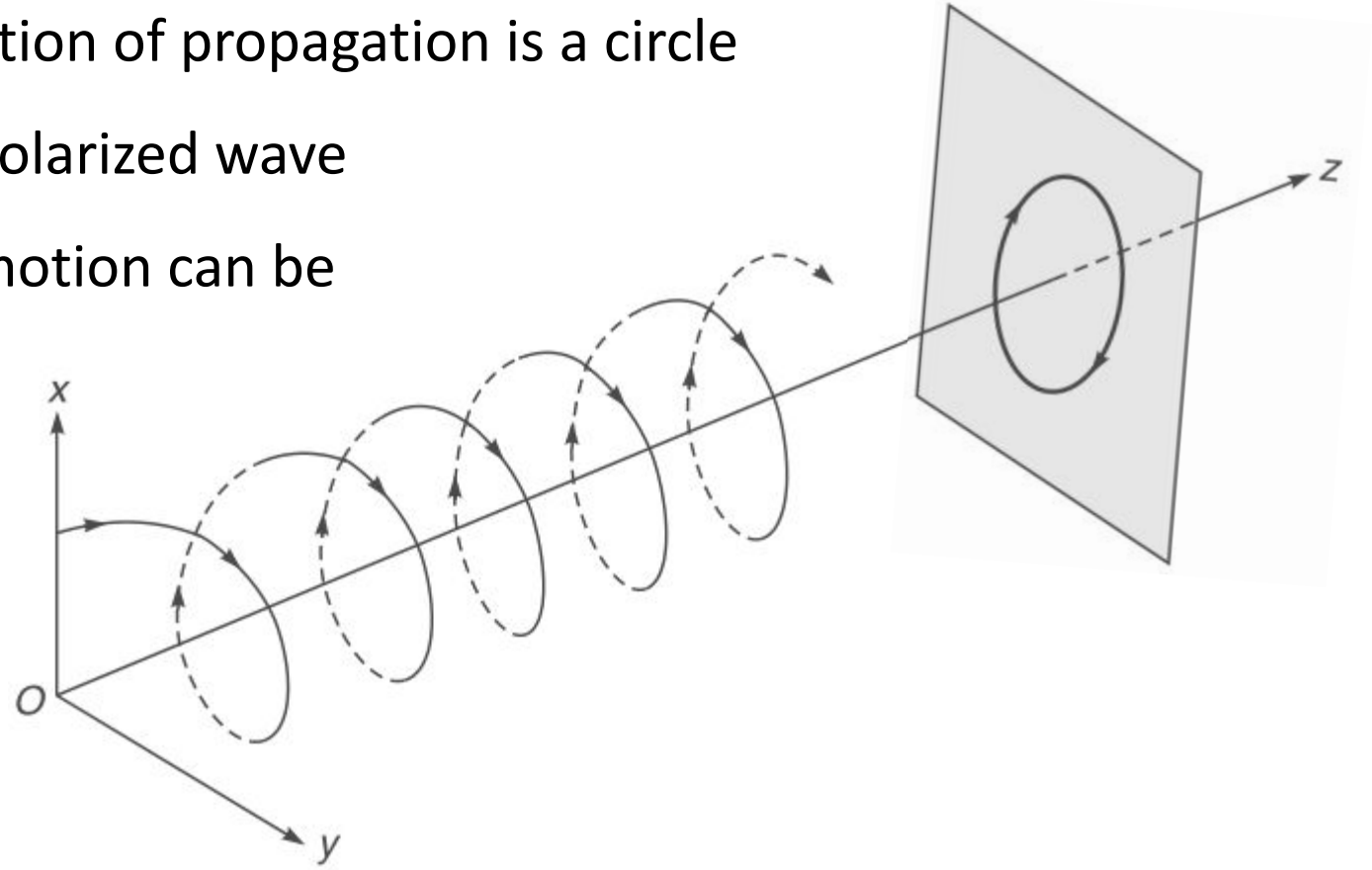
Polarization - analogy to transverse wave on a rope

- If the wave is generated on a rope by oscillating one end in a horizontal plane, vibration exists only in one plane - the horizontal plane
- Tip of the amplitude (of horizontal displacement) vector will trace a horizontal line on a plane perpendicular to the direction of propagation
- This is an example of horizontally linearly polarized wave



Polarization - analogy to transverse wave on a rope

- If the wave is generated on a rope by moving one end in a circular path on a transverse plane, the direction of vibration will continuously change in time
- Tip of the amplitude (of vibration) vector traces a spiral path, whose projection on a plane perpendicular to the direction of propagation is a circle
- This is an example of circularly polarized wave
- We know that uniform circular motion can be obtained from superposition of two orthogonal SHM of same frequency and same amplitude, with a phase difference of $\pi/2$



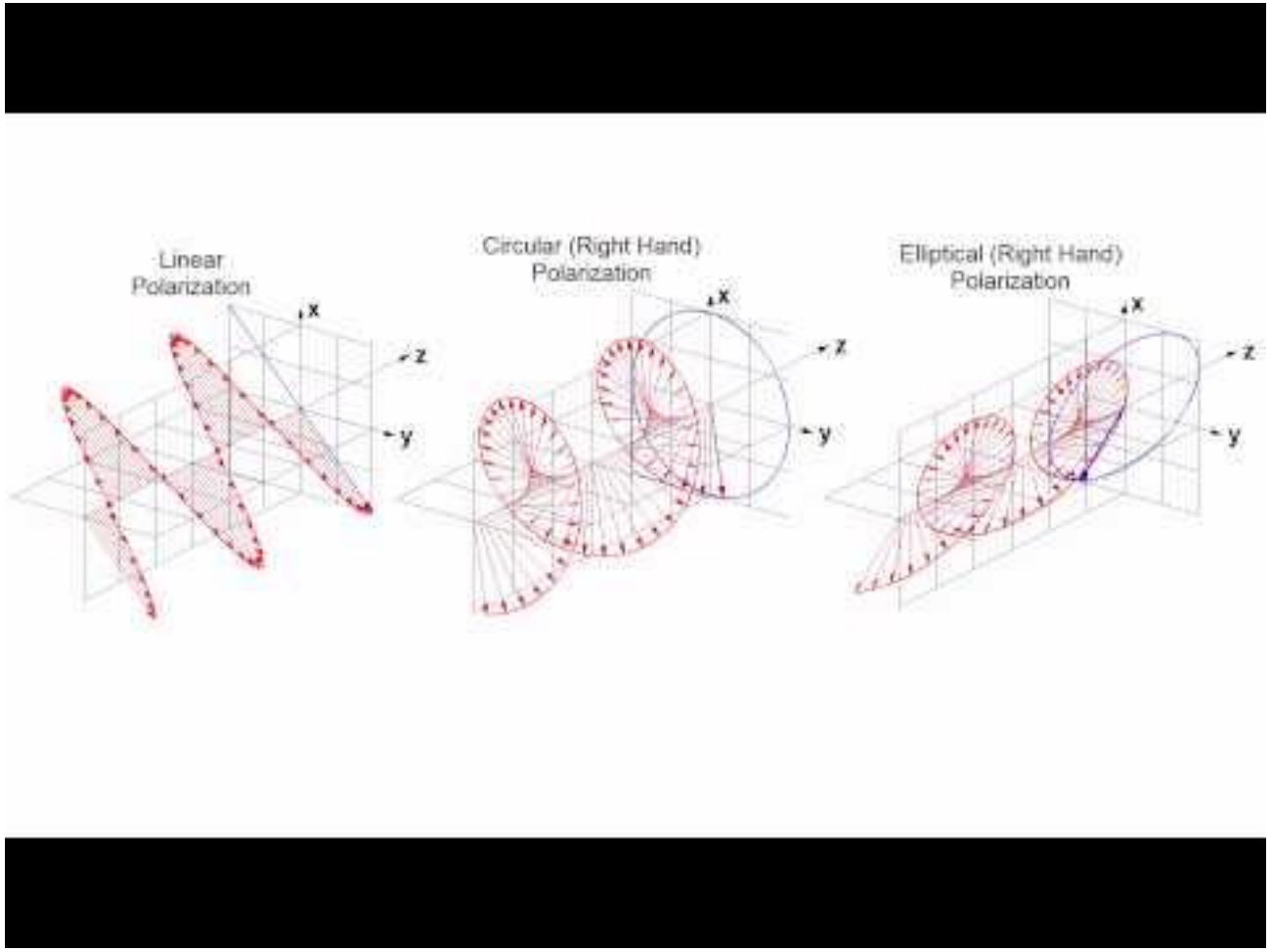
Circularly polarized transverse wave on a rope

https://www.youtube.com/watch?v=5uXjYj72dCY

Elliptic polarization

- If the wave is generated on a rope by moving one end in an elliptic path on a transverse plane, the direction of vibration will continuously change in time
- Tip of the amplitude (of vibration) vector traces a elliptic-spiral path, whose projection on a plane perpendicular to the direction of propagation is an ellipse
- This is an example of elliptically polarized wave - it is more general form of polarization
- Circle or line can be thought of as special cases of ellipse

Animation of linear, circular and elliptic polarization



https://www.youtube.com/watch?v=Q0qrU4nprB0

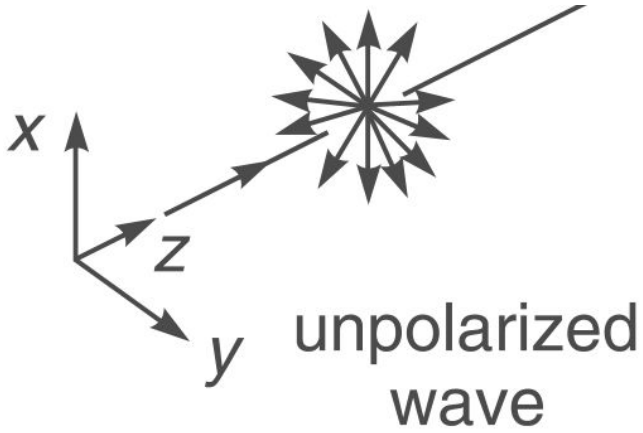
Natural light is unpolarized

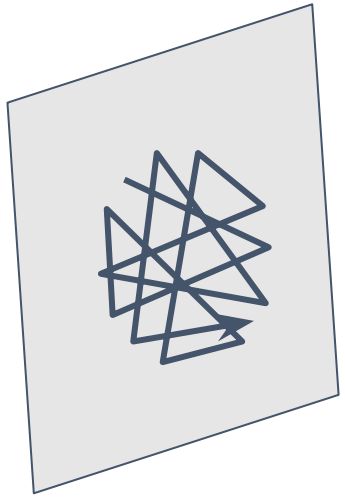
- Light from all natural sources (such as sun, light bulb, candle flame etc.) do not usually have a fixed polarization
- They give out predominantly unpolarized light
- Laser light can be predominantly polarized

Unpolarized wave (randomly polarized wave)

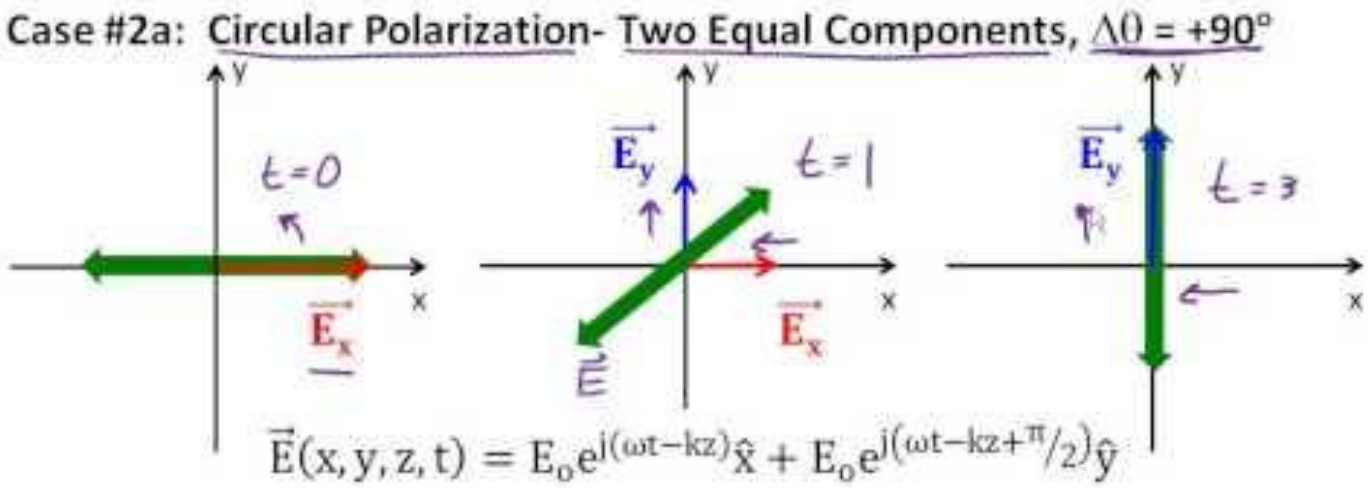
- In all previous example, tip of the amplitude vector was moving in a well-defined, predictive trajectory in time. Different shape of the trajectory was obtained for different polarization of wave
- Imagine a wave on a rope is generated by oscillating one end, but the direction of oscillation was **changed randomly over a short time interval**. Plane of vibration will change randomly and the tip of the amplitude vector will trace a random (unpredictable) trajectory which will fill up the entire transverse plane in some time

__Short compared to the observation time but long compared to time-period of the wave__



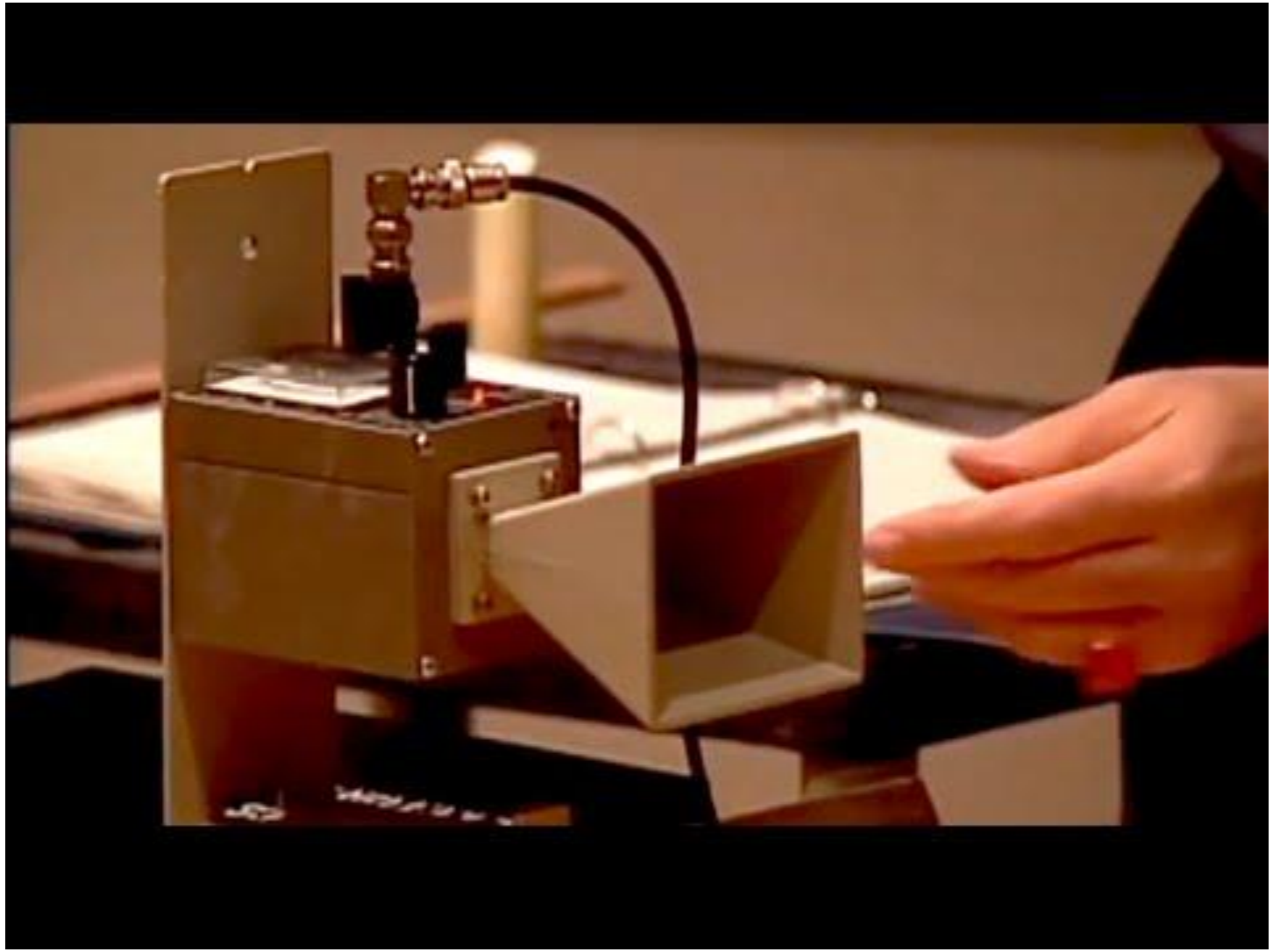


Lectures on polarization of E M waves



https://www.youtube.com/watch?v=84AcKIcF7VY&t=302s

Lectures on polarization of E M waves



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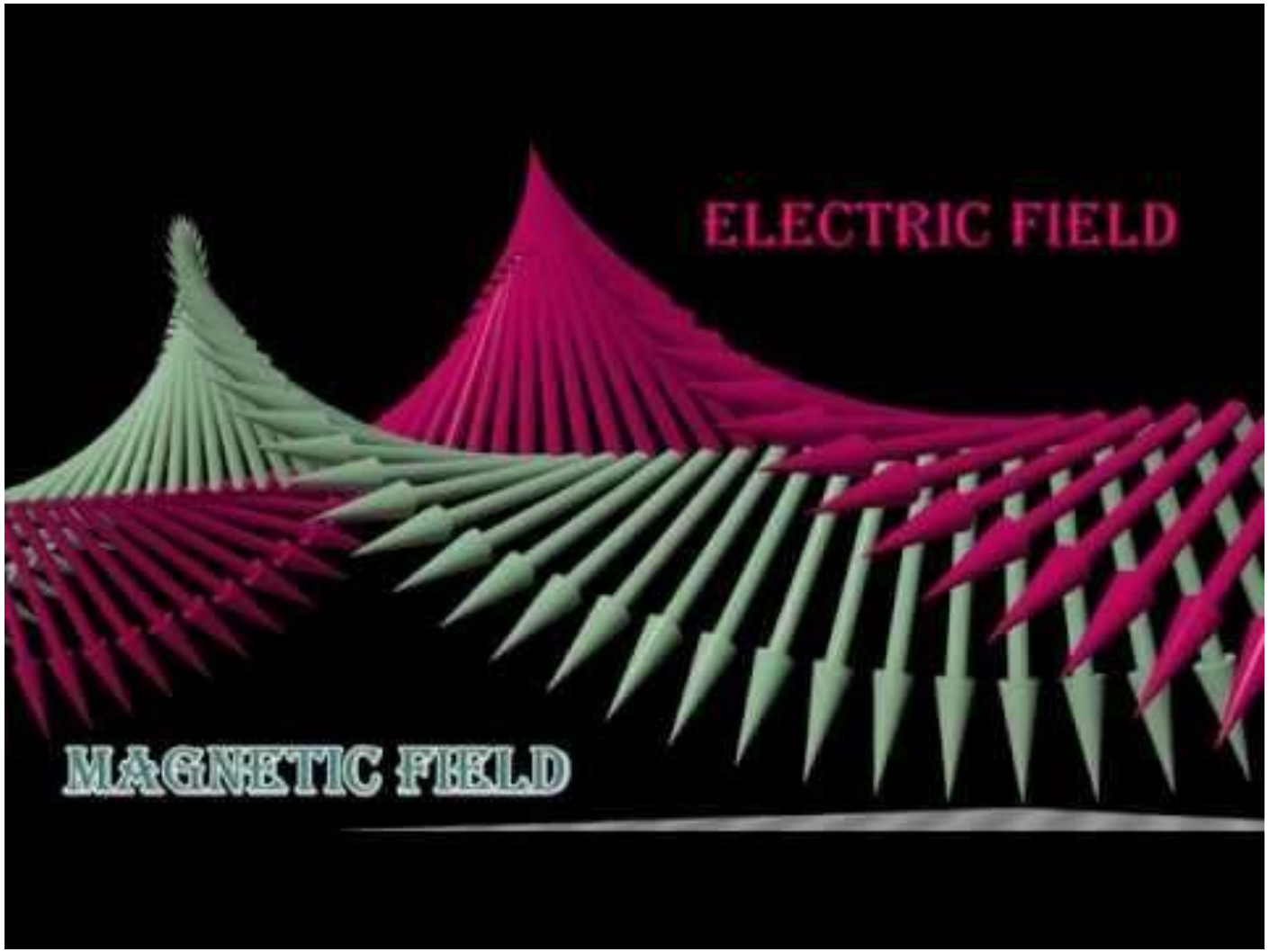
Imaginative visualization of polarization of E M wave



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Polarization of plane E M waves

- Two orthogonal linear polarizations
- Linear polarized in x-direction:

$$\vec{E} = \hat{x} E_{0x} \cos(\omega t - kz + \phi_x)$$

Corresponding B-field:

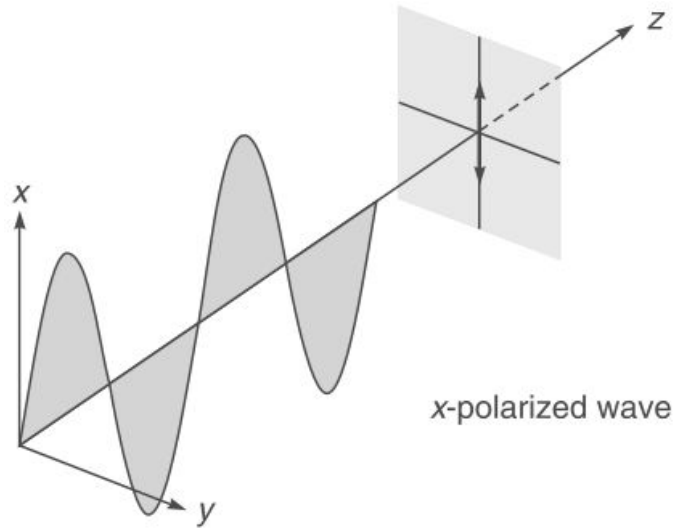
$$\vec{B} = -\hat{y} \frac{E_{0x}}{c} \cos(\omega t - kz + \phi_x)$$

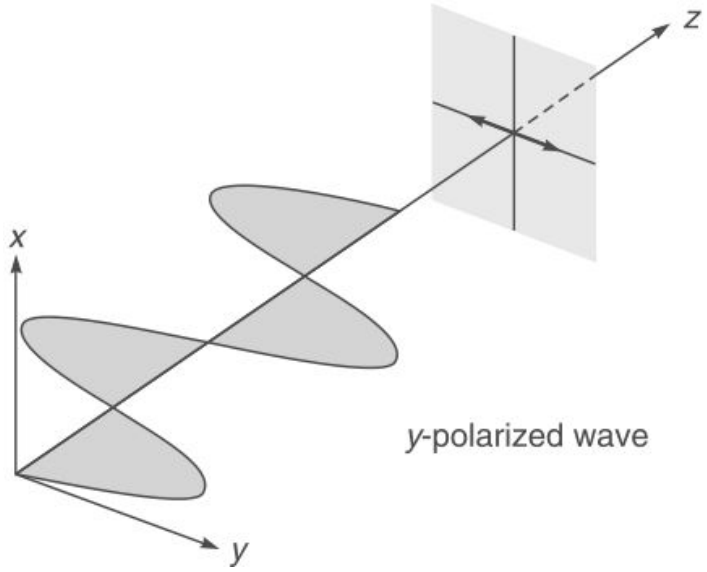
- Linear polarized in y-direction:

$$\vec{E} = \hat{y} E_{0y} \cos(\omega t - kz + \phi_y)$$

Corresponding B-field:

$$\vec{B} = \hat{x} \frac{E_{0y}}{c} \cos(\omega t - kz + \phi_y)$$



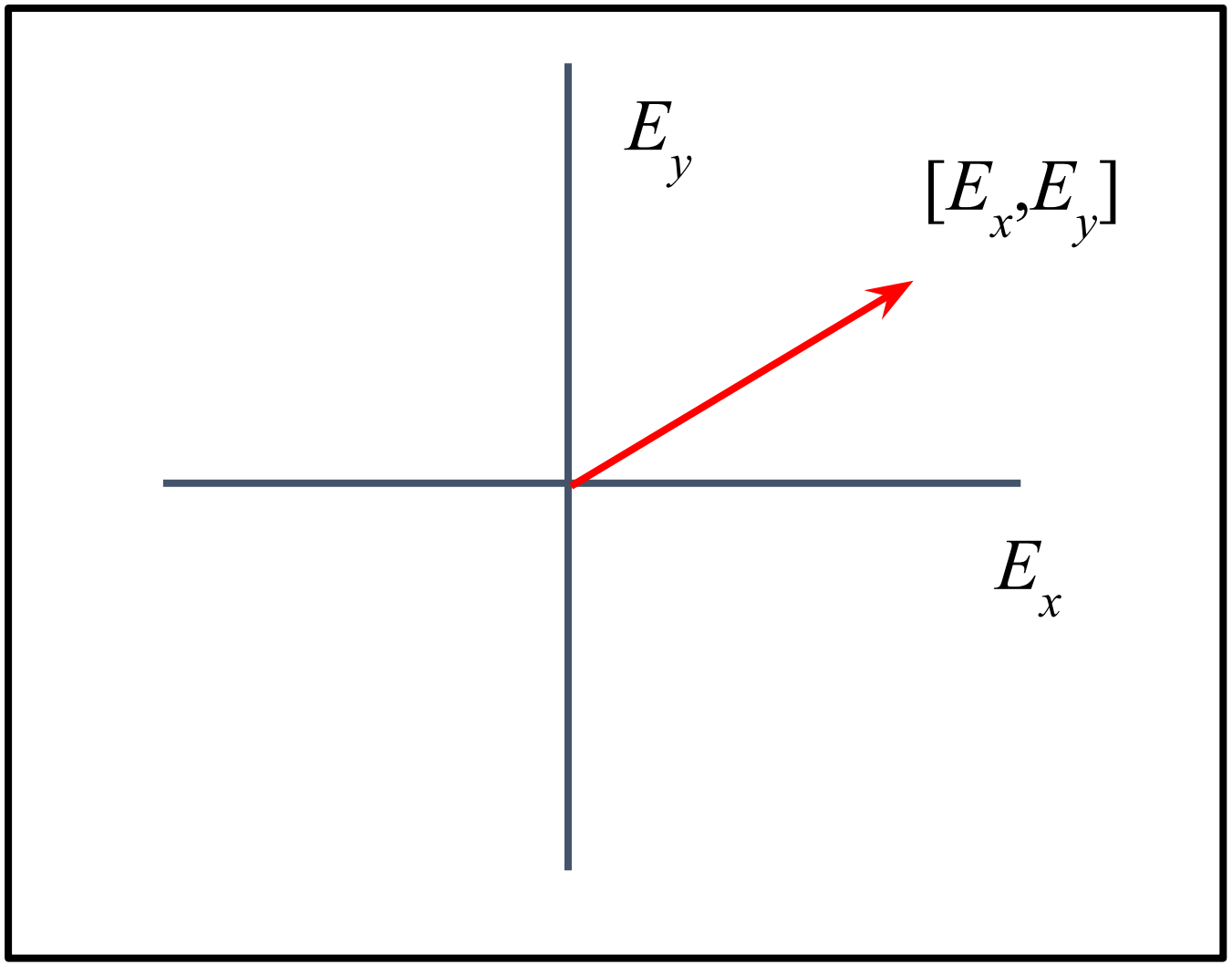


Two orthogonal linear polarization as basis set

- Linearly x-polarized and y-polarized waves are orthogonal to each other and are independent of each other $\longrightarrow$ They can exist simultaneously
- They can form an ideal basis set to express any arbitrary polarization i.e. any arbitrary polarization of E M wave can be expressed as a linear superposition of these two polarization state
- There can be other orthogonal polarizations giving other basis sets. For example,
  - linearly polarized at 45° and 135°
  - right and left circularly polarized

Two orthogonal linear polarization as basis set

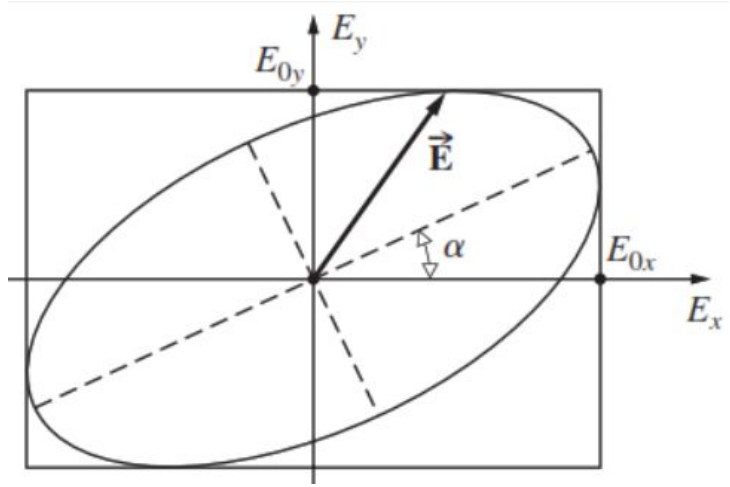
- Any arbitrary polarization of E M wave can be expressed as a linear superposition of linearly x- and y-polarized states
- $\vec{E}(t,z) = \hat{x}E_{0x}\cos(\omega t - kz + \phi_x) + \hat{y}E_{0y}\cos(\omega t - kz + \phi_y)$
- We are interested in the trajectory of the tip of the E-vector on a fixed plane perpendicular to the direction of propagation. We can take the fixed plane to be at $z = 0$
- $\vec{E}(t,z=0) = \hat{x}E_{0x}\cos(\omega t + \phi_x) + \hat{y}E_{0y}\cos(\omega t + \phi_y) = \hat{x}E_x(t) + \hat{y}E_y(t)$
- To get the trajectory of the tip of E-vector, we have to follow the time-evolution of the point $[E_x(t), E_y(t)]$ in the E_x-E_y plane


E_y
$[E_x, E_y]$
E_x

Superposition of two orthogonal SHM

- $$\vec{E}(t, z=0) = \hat{x}E_{0x}\cos(\omega t) + \hat{y}E_{0y}\cos(\omega t + \phi)$$
- Trajectory of tip of E-vector is given by

$$\left(\frac{E_x}{E_{0x}}\right)^2 + \left(\frac{E_y}{E_{0y}}\right)^2 - 2\left(\frac{E_x}{E_{0x}}\right)\left(\frac{E_y}{E_{0y}}\right)\cos\phi = \sin^2\phi$$

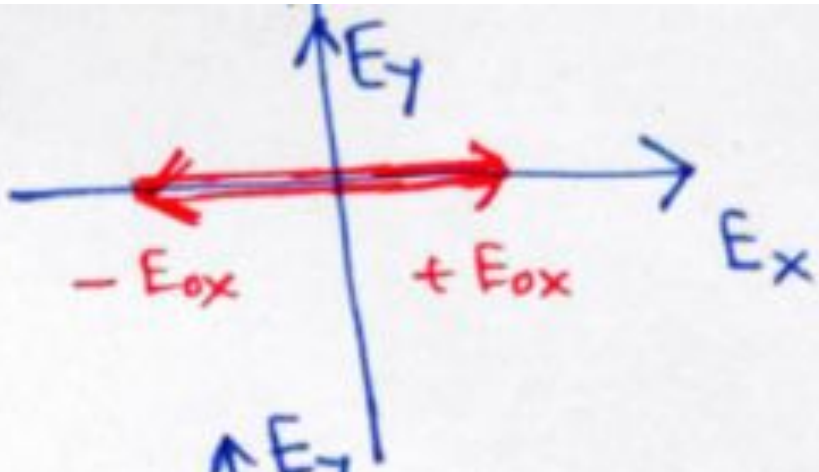


- This describes an ellipse whose major axis is rotated with respect to E_x axis in the E_x-E_y plane by angle α which is given by $\tan 2\alpha = \dfrac{2E_{0x}E_{0y}\cos\phi}{E_{0x}^2 - E_{0y}^2}$

A few more cases of interests

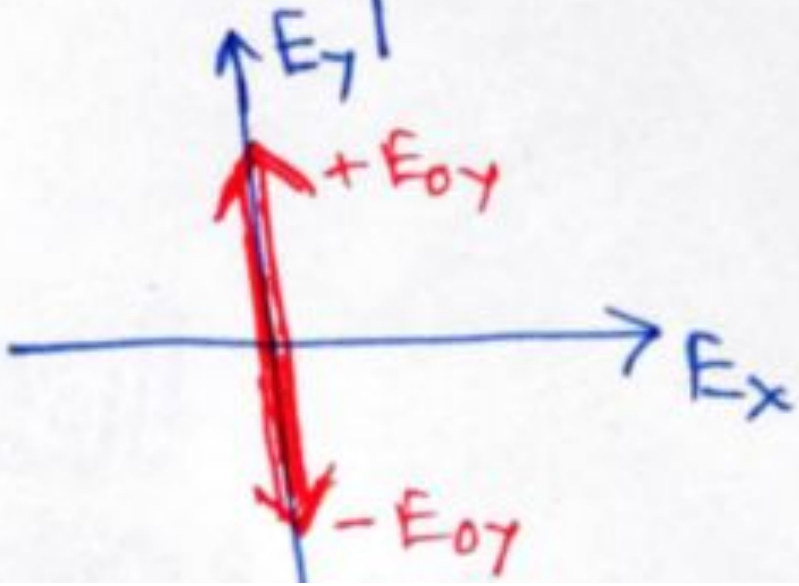
① $E_{0x} \neq 0,\ E_{0y} = 0$

Linear x-polarized light



② $E_{0y} \neq 0,\ E_{0x} = 0$

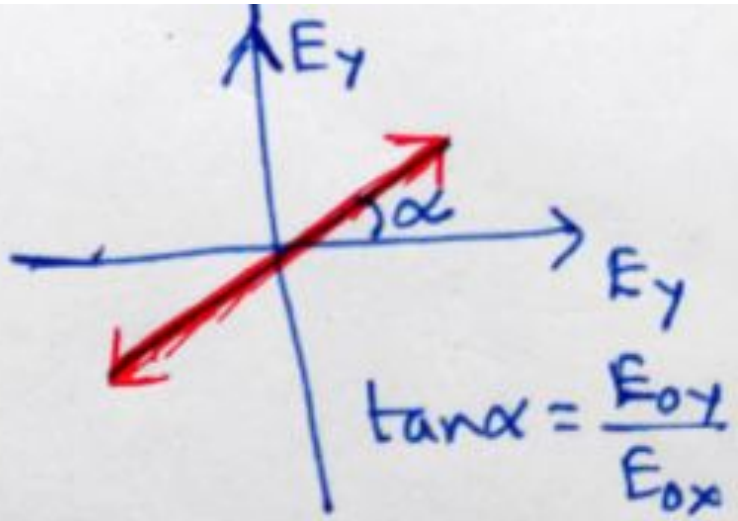
Linear y-polarized light



A few more cases of interests

③ $E_{0x} \neq E_{0y} \neq 0, \quad \phi = \phi_x - \phi_y = 0$

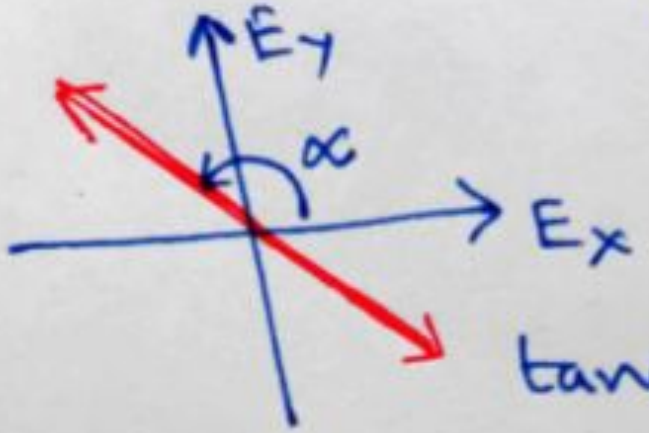
Linearly polarized at angle α with respect to E_x direction



$$\tan\alpha = \frac{E_{0y}}{E_{0x}}$$

④ $E_{0x} \neq E_{0y} \neq 0, \quad \phi = \pm\pi$

Linearly polarized at angle α with respect to E_x direction



$$\tan\alpha = -\frac{E_{0y}}{E_{0x}}$$

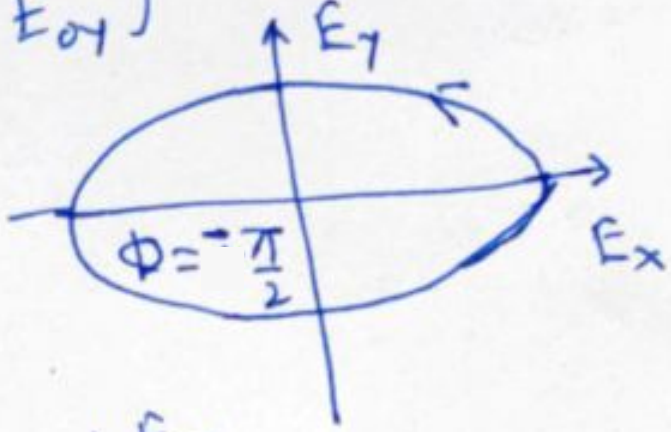
A few more cases of interests

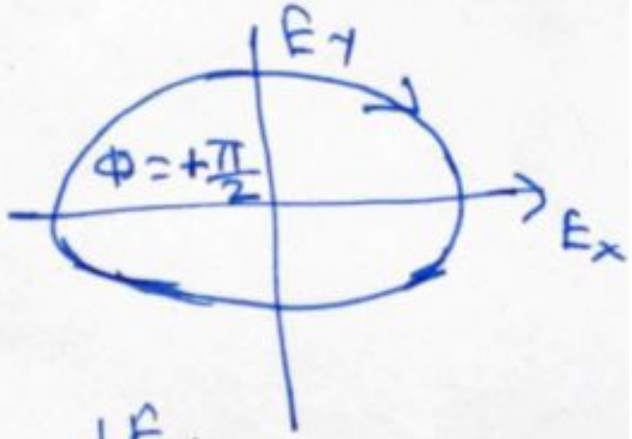
⑤ $\underline{\phi = \pm \frac{\pi}{2}}$: $E_x = E_{0x} \cos(\omega t)$, $E_y = E_{0y} \cos(\omega t \pm \frac{\pi}{2})$

$= \mp E_{0y} \sin(\omega t)$

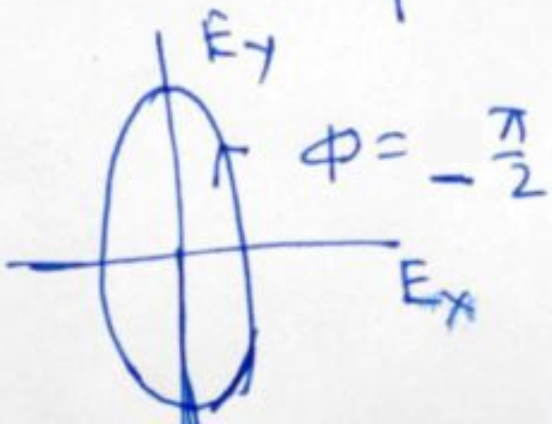
$$\Rightarrow \left(\frac{E_x}{E_{0x}}\right)^2 + \left(\frac{E_y}{E_{0y}}\right)^2 = 1$$

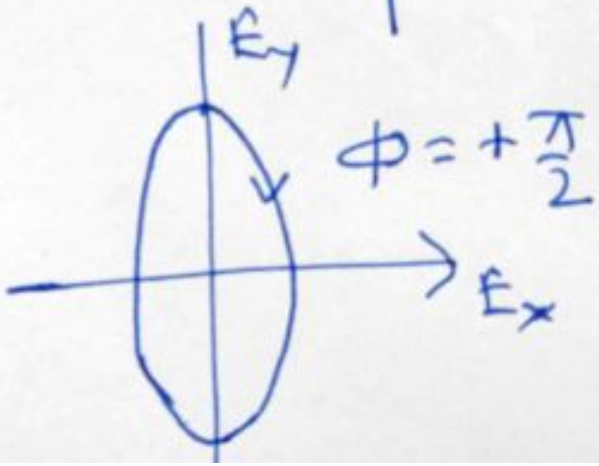
a) $\underline{E_{0x} > E_{0y}}$





b) $\underline{E_{0y} > E_{0x}}$

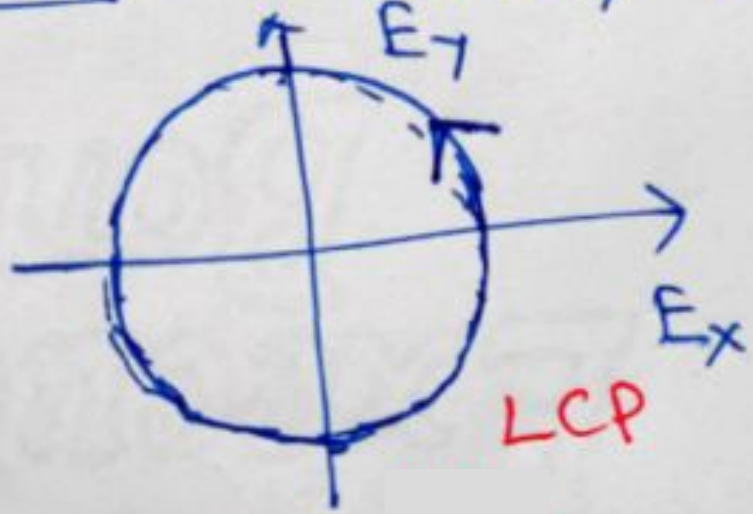




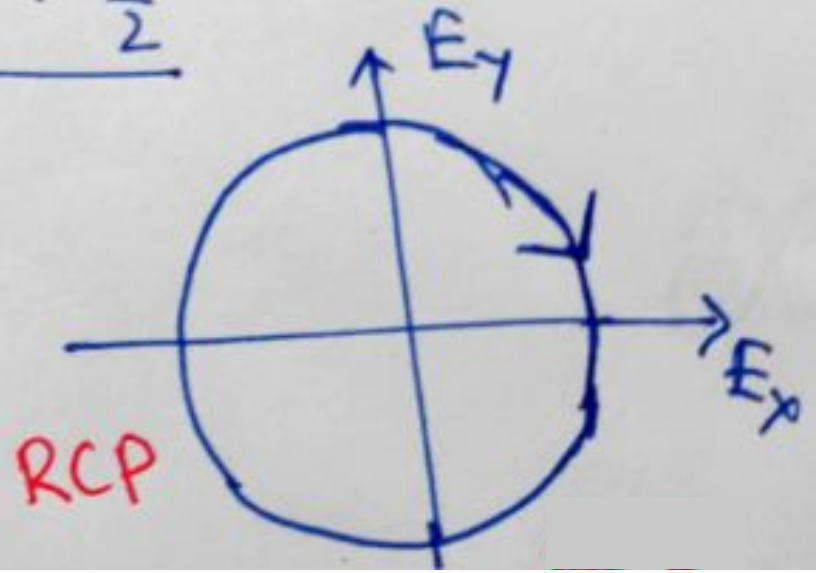
A few more cases of interests

c) $E_{0x} = E_{0y}$: Circularly polarized light

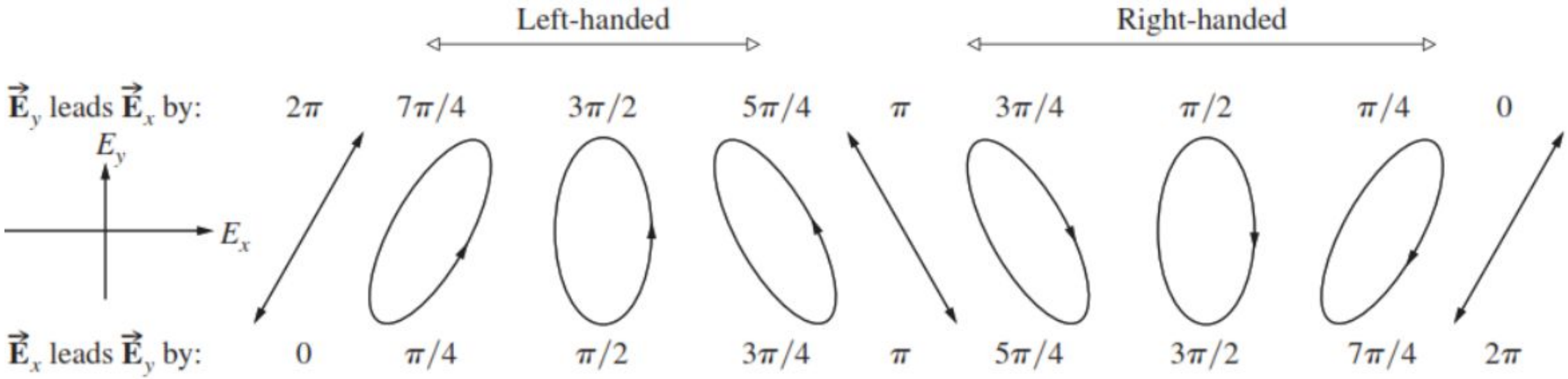
$\phi = -\frac{\pi}{2}$



$\phi = +\frac{\pi}{2}$



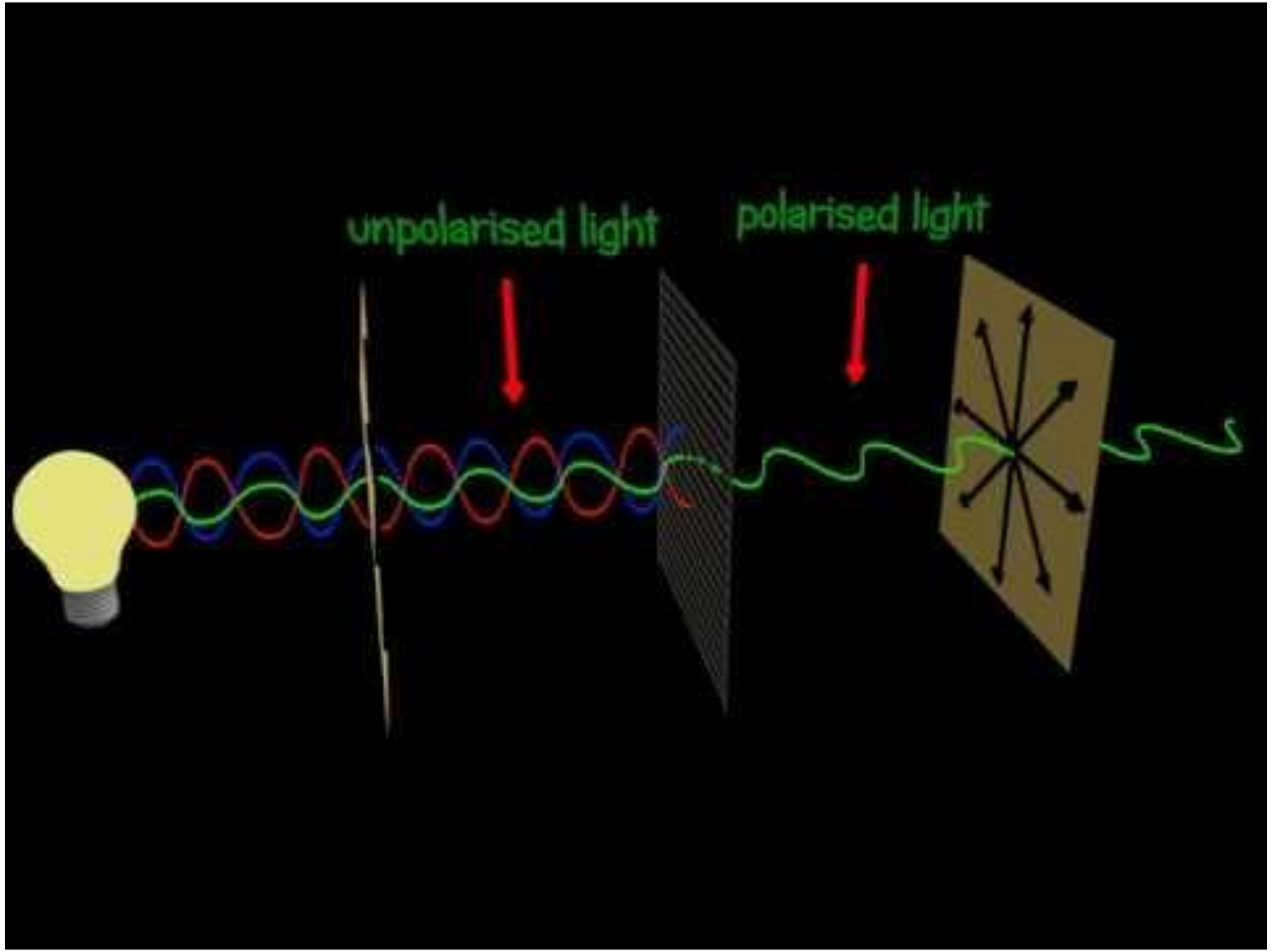
Elliptical polarization for different phase difference ϕ



Unpolarized light

https://www.youtube.com/watch?v=m8hQ2un6nqw

Unpolarized (randomly polarized) waves



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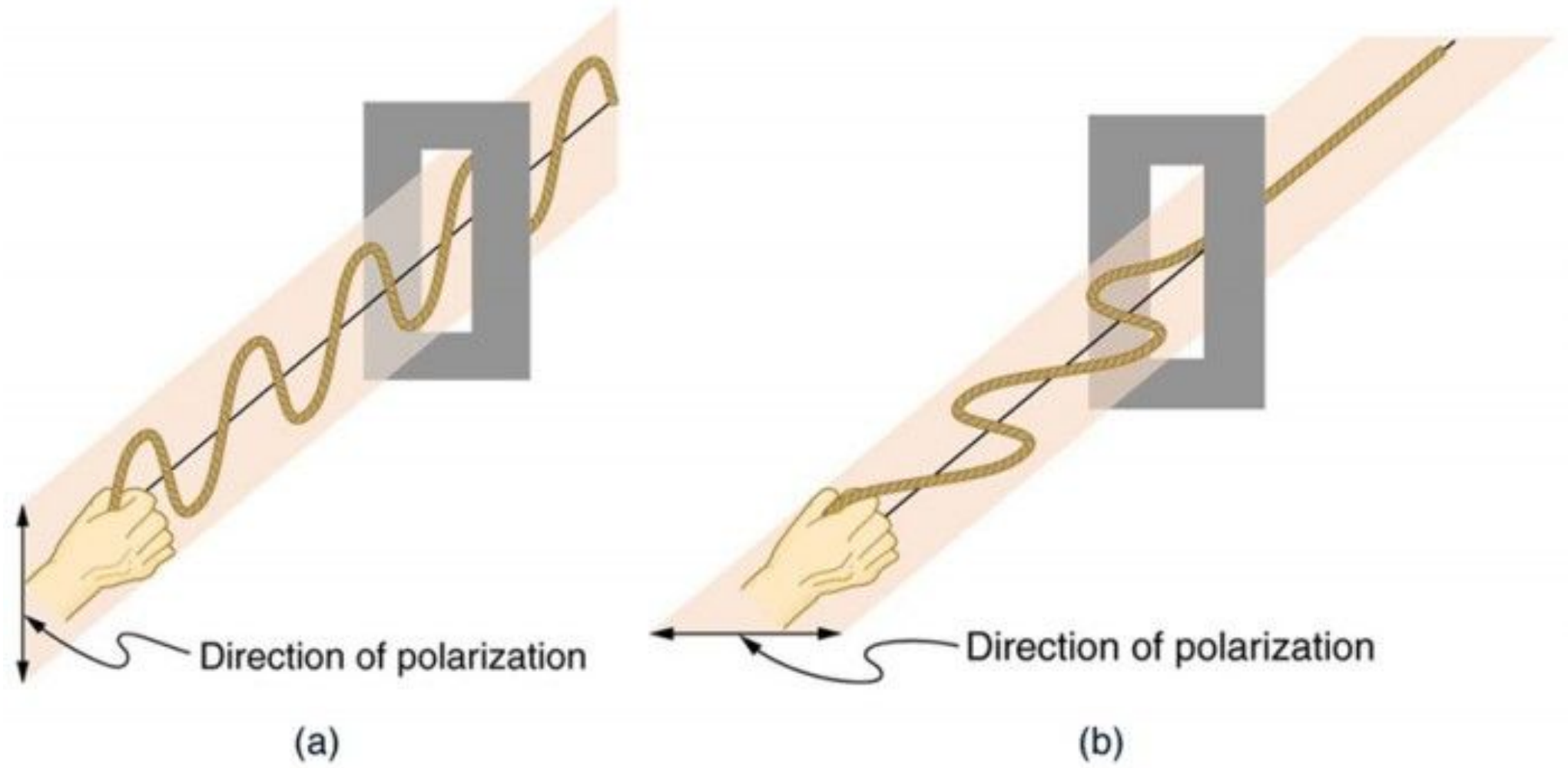
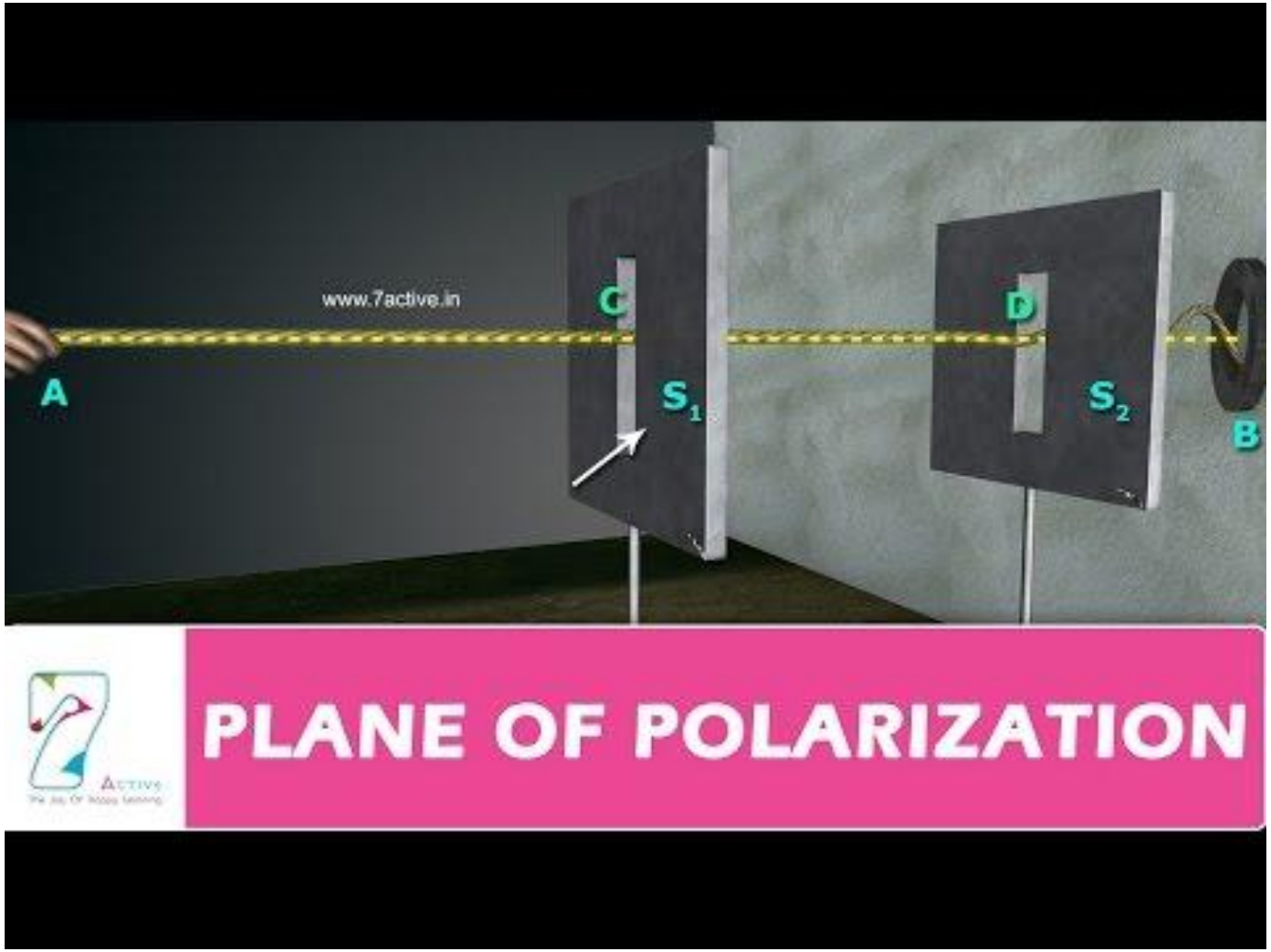
Polarization filtering



- Vertical polarization filter allows vertical oscillations to pass through but it blocks horizontal oscillations

Polarization filtering

- Vertically polarized wave passes through a vertical polarization filter but it is blocked by a horizontal polarization filter



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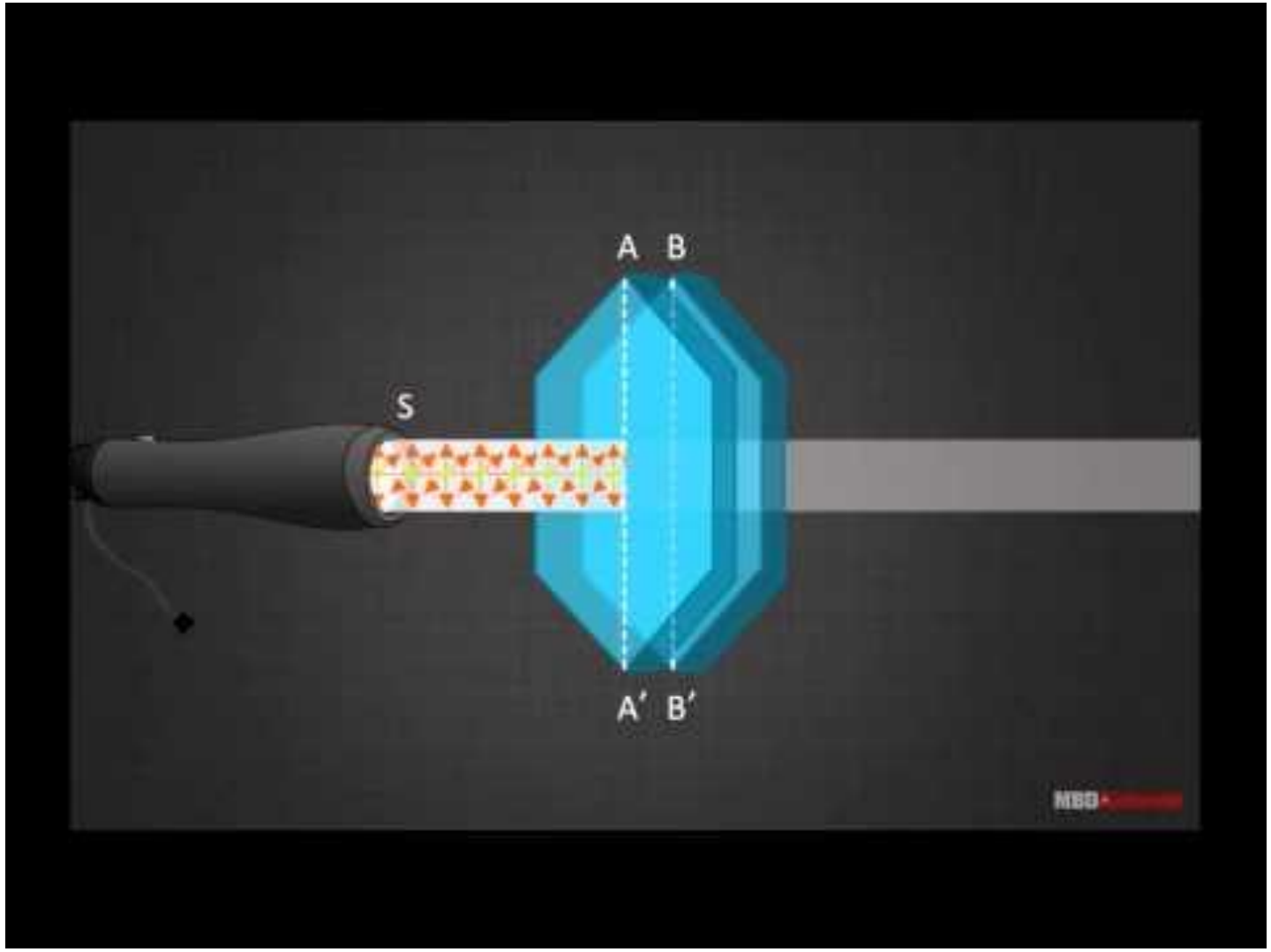
Polarization filtering



https://www.youtube.com/watch?v=6_C8KyU67RU

Experimental demonstration of Malus's law (1809)

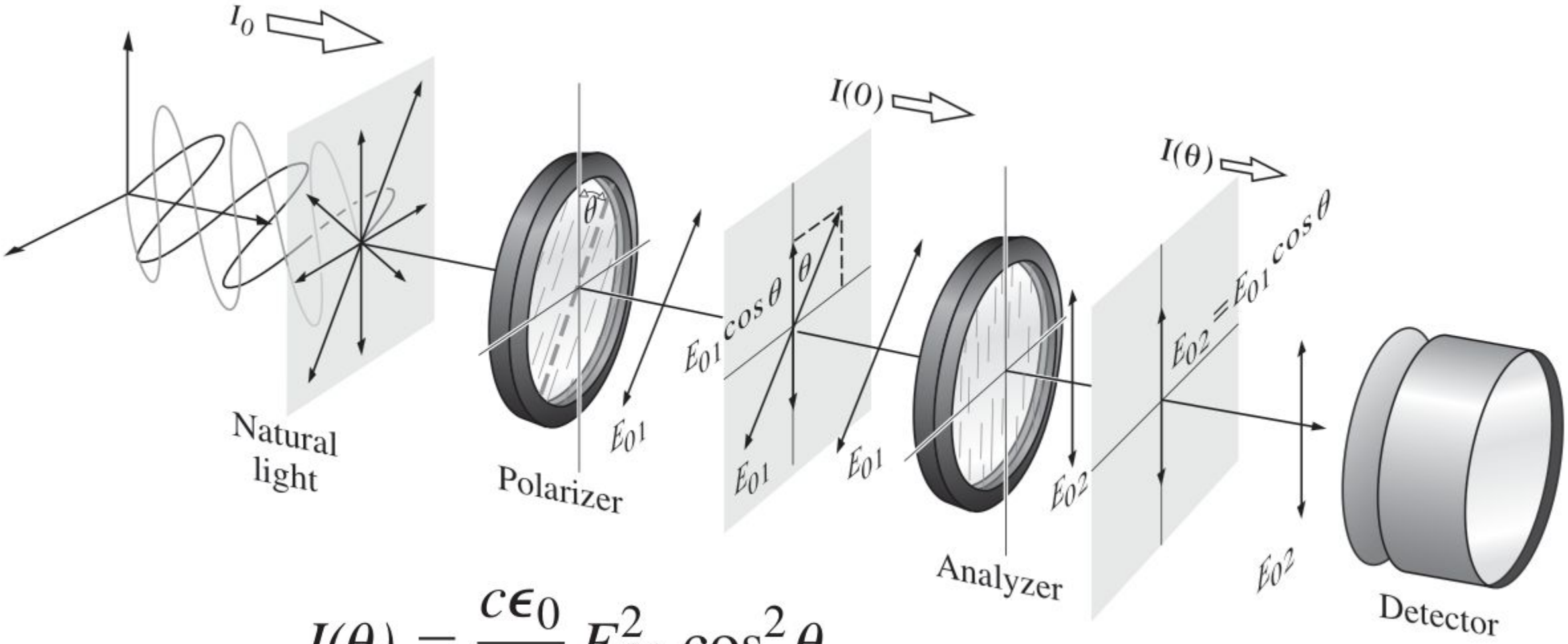
Étienne-Louis Malus

Étienne-Louis Malus

Born	23 July 1775 Paris
Died	24 February 1812 (aged 36) Paris
Nationality	French
Known for	Polarization of light Malus's law
Awards	Rumford Medal (1810)
Scientific career	
Fields	Physics

https://www.youtube.com/watch?v=hBJ6GSvPg2M

Malus's law: $I(\theta) = I(0)\cos^2\theta$



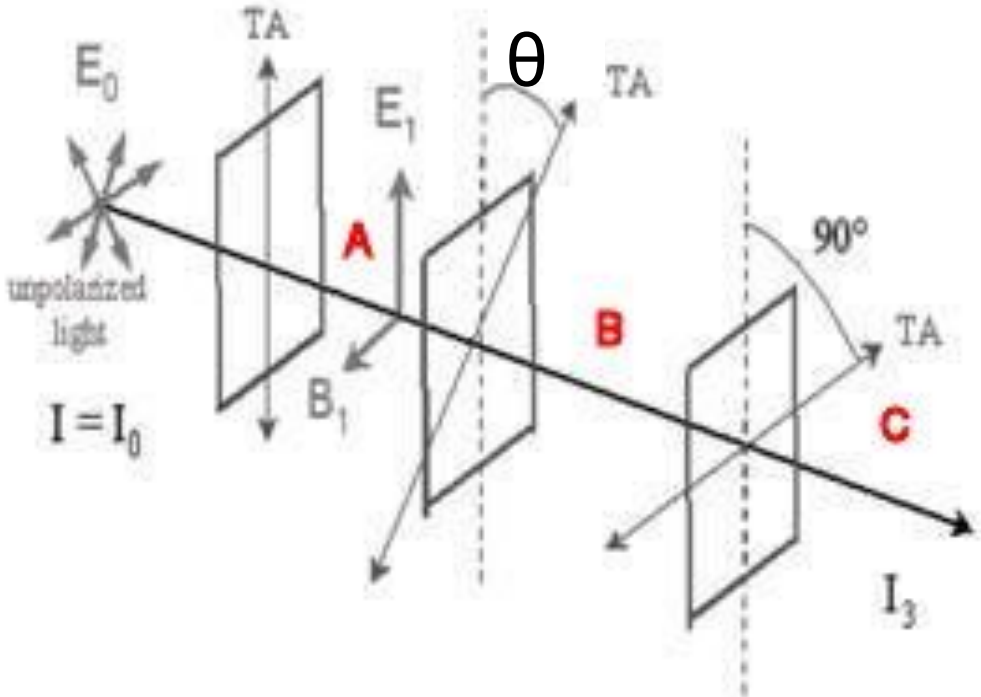
$$I(\theta) = \frac{c\epsilon_0}{2} E_{01}^2 \cos^2\theta$$

Key concepts from Malus's law

1. When unpolarized light passes through a polarizer, the intensity is reduced by a factor of ½. The transmitted light is polarized along the axis of the polarizer. $I = I_0 \langle \cos^2\theta \rangle = I_0/2$
2. When polarized light of intensity I_0 is incident on a polarizer, the transmitted intensity is given by $I = I_0 \cos^2\theta$, where θ is the angle between the polarization direction of the incident light and the axis of the polarizer. The transmitted light is polarized along the axis of the polarizer.

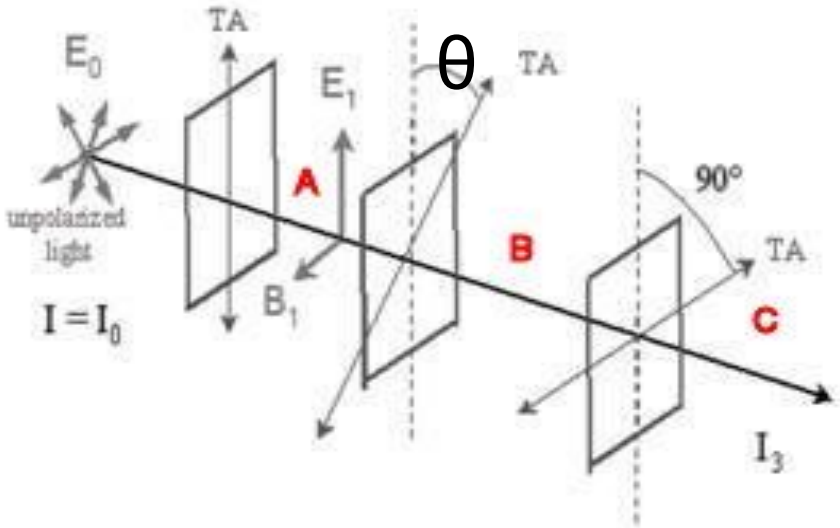
Calculation involving 3 polarizers

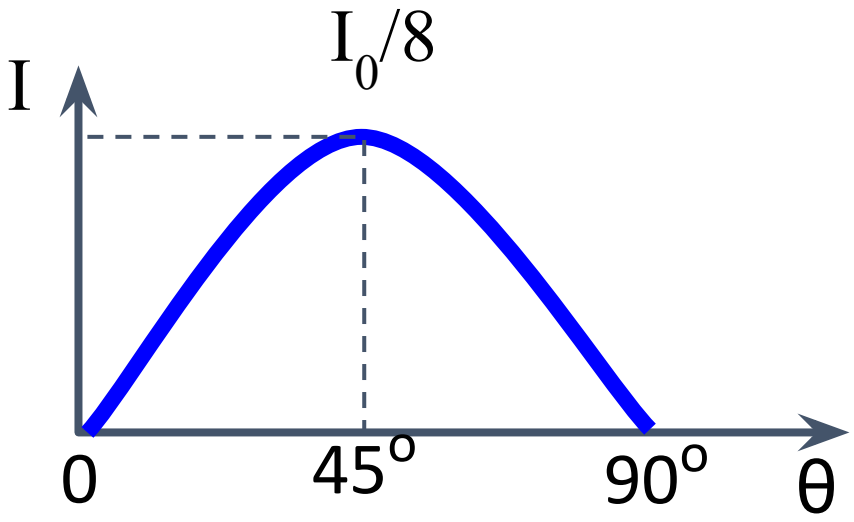
Unpolarized light is sent through 3 polarizers. 1st and 3rd polarizers are in a crossed configuration. The middle polarizer is making an angle θ with the 1st polarizer (vertical) axis. Write down the intensity in the region C as a function of θ and plot it schematically ($0^o \leq \theta \leq 90^o$). What angle of the middle polarizer maximizes the intensity in region C?



Solution:

1. Region A: Intensity is $I_0/2$, light is vertically polarized.
2. Region B: Intensity is $(I_0/2)\cos^2\theta$, light is polarized along 2nd polarizer axis.
3. Region C: Intensity is $(I_0/2)\cos^2\theta\cos^2(90^o - \theta) = (I_0/8)\sin^2(2\theta)$, light is horizontally polarized.
4. Intensity in region C will reach a maximum of $I_0/8$ for $\theta = 45^o$





Quarter wave plate

1. QWP can be used to convert
- Linear to circular polarization
- Circular to linear polarization
- Elliptical to linear polarization

2. Unpolarized light remains unpolarized after passing through a QWP

3. If light incident normally on a QWP is linearly polarized along its optic axis or perpendicular to it, then there will be no change in the polarization state

https://www.youtube.com/watch?v=EBVNbRN805o

Polarization of white (non-monochromatic or polychromatic) light

https://www.youtube.com/watch?v=2c7VIg8-LVI

How to rotate plane of polarization by using a QWP and a polarizer

- Convert the linearly polarized light to circularly polarized light by using a QWP.
- Then you can directly set the polarizer after QWP to the desired angle at which you need to have the axis of linear polarization

https://www.youtube.com/watch?v=mNFRaM-2cvg&list=PL4E7FAAD67B171EBC&index=9